

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI  
II SEMESTER 2016-2017  
EEE/CS/INSTR F241 MICROPROCESSOR PROGRAMMING AND INTERFACING  
Comprehensive Exam (Closed BOOK)

M.M: 40

13-05-2017

DURATION: 50min

- Note 1. Write your final answer in the separate answer sheet provided  
2. Use main answer sheet for rough work

Q1. Arrange the following interrupts in 8086 microprocessor in **increasing** order of their priority:

A. INTO                      B. Single Step                      C. NMI                      D. INTR

[2M]

Q2. Write down the **addressing mode** of the following instruction for an 8086 processor

MOV AX, 4020[BX+DI]

[3M]

Q3. An 8086 processor is working with 10 MHz system clock, how much **time** (with proper units) one MEMR cycles takes if there are 4 wait states. [3M]

Q4. For an 8254 operating in mode-3 with input clock frequency of 2 MHz if the counter is loaded with 0000H then the output clock frequency is? (report up-to two decimals) [4M]

Q5. Determine the contents of **AL** register ( **in hex**) after completing execution of the following program snippet on 8086 processor. [4M]

MOV AL, 22H

MOV CL, 03H

ROL CL, CL

ROL AL, CL

Q6. Using **only logical** instructions (use minimum no of instruction) from 80286 instruction set, sign extend a 12-bit data in AX register into 16-bit data. Initially the upper nibble of AH is all zeros. [4M]

Q7. In 8086 write a sequence of four instructions that will set all the conditional flags and reset all the status flags. The flag register of 8086 is shown below. [4M]

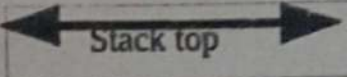
|    |    |    |    |    |    |    |    |    |    |   |     |   |   |   |    |
|----|----|----|----|----|----|----|----|----|----|---|-----|---|---|---|----|
| 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5 | 4   | 3 | 2 | 1 | 0  |
| X  | X  | X  | X  | OF | DF | IF | TF | SF | ZF | X | ACF | X | P | X | CY |
|    |    |    |    | 1  | 1  | 0  | 0  | 1  | 1  |   | 1   |   | 1 |   | 1  |

Q8. In 8086 the flag register is a 16-bit register in which the least significant bit of the upper byte corresponds to trap flag. Write a sequence of instructions (code snippet) which can set the trap flag while leaving the other flags unchanged. [4M]

**Q9.** Write an code snippet (need now show instruction for segment initialization) to generate a delay of 63ms using an 8086 processor running at 5MHz frequency. Use only instruction from the table given below: [4M]

| Instruction | No. of T-states needed |
|-------------|------------------------|
| MOV         | 4                      |
| NOP         | 3                      |
| DEC/INC     | 2                      |
| JNZ         | 16                     |

**Q10.** An ISR is being is executed by 8086 and the contents of the stack at that time are shown below. If the ISR doesn't modify the contents of the stack during its execution then the return address (20-bit Hex) of the ISR is : [4M]

|     |                                                                                     |
|-----|-------------------------------------------------------------------------------------|
| 00H |  |
| 65H |                                                                                     |
| 00H |                                                                                     |
| 14H |                                                                                     |
| 00H |                                                                                     |
| 15H |                                                                                     |
| 08H |                                                                                     |
| 16H |                                                                                     |

**Q11.** Write the values of the destination after the execution of the code snippet listed below, write answer in hexadecimal only [4M]

```

STC
MOV AX, 5485H      ; AX =
RCR AL, 1          ; AL =
MOV CL, 03         ; CL =
RCL AX, CL         ; AX =
  
```



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**Comprehensive Exam (OPEN BOOK)**

**M.M:80**

**13-05-2017**

**DURATION: 2hrs**

**Q1.** Design a 80386 based microprocessor system with 24-bit address bus and 16 bit data bus with 7MB of memory requirement starting at 000000H address. This 7MB is divided into 2MB of SRAM1 at the bottom then 4MB of SRAM2 at the centre and 1MB of ROM at the top. Following components are provided in required number : 512KB SRAM1 chips, 1MB SRAM2 chips, 512KB of ROM chips.

- Write the entire range (memory map) of RAM/ROM addresses in hexadecimal only.
- Now, draw the complete labelled memory interfacing diagram using absolute addressing using only single 74LS138 decoder (3:8), 2 Input NAND gate-2 Nos and minimum number of NOT gate. In your schematic you need to show address buses, data buses and chip select logic and Read, write signals. Assume BHE', MEMR', MEMW' are readily available for interfacing. (Note: Label all pins and wires in your design) **[6+12=18M]**

**Q2.** If the root directory entry of a file is the following:

42 49 54 53 44 41 54 41- 58 4C 53 23 7B BD 4D 18  
A4 3A 4A 3A 00 00 0F 20- CE 28 0F 00 04 00 00 04

- What are the attributes of this file?
- What is the date of creation of this file in Date.Month.Year format ?
- What is the time of creation of this file in Hour:Min:Sec format?
- What is the starting cluster number of this file?
- What is the size of this file in bytes ? **(2+2+3+2+2=10M)**

**Q3.** An 8086 system has a master 8259A and two slave 8259As. The master 8259A should be I/O mapped such that it is programmable using port addresses F220H and F222H. Slave-1 should be mapped to addresses F230H and F232H and should be connected to IR7 of the master. Slave-2 should be mapped to addresses F240H and F242H and should be connected to IR6 of the master.

- Draw a schematic which shows interfacing of an 8086 with these 8259A chips for above mentioned scheme. Use a single 74LS138 decoder and minimum logic gates for decoding.
- The 8259A's are setup for edge triggered interrupts with the slave-1 interrupt type as F0H with IR0 through IR7 enabled. The Slave-2 interrupt type is F8H with IR0 and IR1 enabled. The master 8259A generates an interrupt type number of 28H. Both master and slave devices are configured in buffered mode, no special fully nested mode and the IR inputs of the master are masked except IR6 and IR7. All should use AEOI and default priority settings. Now, write the required ICWs and OCWs (in hex) in a table format corresponding to each 8259. **(18M)**

**Q4.** There are four teams in a quiz contest. In the "buzzer" round of the contest, any team which wants to answer a question posed by the quiz-master has to press a button "ANSW" placed their desk. Each team has their own ANSW button. Also whenever a team presses ANSW, a small alarm or buzzer which is placed on each team's table rings for 0.5 seconds to indicate which team has pressed ANSW button. A single press of this button produces a high going pulse which stays at logic-1 for 0.5 seconds before returning to logic-0. After pressing this "ANS" button, that particular team gets 10 seconds to answer their question (excluding the 0.5 Sec for which pulse remains high). The counting of this 10 second duration begins only when a team presses their ANSW button. There a single LED on the desk of the quiz-master which glows for 1 sec to indicate that the 10-second duration given to a team has



elapsed. The quiz-master also has switch called "ENL" on his desk which is used to enable the above mentioned LED. This LED glows only when ENL is enabled and the 10-second duration given to a team has elapsed. So before posing a question to each team, he enables ENL. If a team answers the question correctly before the 10-second duration is completed, the quiz-master disables ENL so that LED does not glow. If a team is unable to answer a question correctly, the quiz-master reveals the correct answer and moves onto the next question.

#### Assumptions:

- No two teams press the button ANSW at the same time
- Once a team presses their button ANSW, the ANSW button of any team is not pressed until next question
- The circuit that produces a high going pulse for 0.5 seconds on the press of button ANSW (as mention above) is already provided.
- IORD', IOWR', MEMRD' and MEMWR' signals are available

Now, implement this 10-second timer system, which counts the time using ANSW buttons using a single 8254. There are no other ICs available except logical gates and resistors, LEDs, switches, etc. The 8254 is programmed only once during initialization in the beginning. The base address of 8254 is 80H. It is an I/O mapped I/O and it has only one clock source of 2MHz.

a. Show the hardware schematic of 8254 for this timer system with address decoding logic using absolute addressing. You have been given: 4 Input OR gates- 6 Nos., NOT gates-5 Nos, AND gate-1 No. Use only the required number of components from the above list.

b. Also, mention the mode in which each counter of 8254 is configured to.

c. Write an ALP code snippet to initialize the 8254 for this design.

(8+6+4=18M)

Q5. For an 80386 microprocessor, a logical memory address described by segment selector:offset is 001Ch:00001000h. Also, GDTR = 000000h, LDTR = 0010h & CR3 = 00001000h. The GDT, LDT, PAGING DIRECTORY and PAGING TABLES are given below.

a. Now, determine the Linear Address generated by the segmentation unit.

b. Also, determine the Physical address generated by the Paging unit.

(16M)

| GDT     | LSB<br>00 01 02 03 04 05 06 07 | MSB |
|---------|--------------------------------|-----|
| 000036h | FE 00 00 01 FF D3 00 00        |     |
| 000030h | FF 01 00 00 20 D2 00 00        |     |
| 000028h | 01 00 00 F0 11 E1 00 00        |     |
| 000020h | A0 00 00 AA 01 D2 00 00        |     |
| 000018h | D0 00 01 01 01 D2 00 00        |     |
| 000010h | FF FF 08 01 00 5F 00 00        |     |
| 000008h | FF FF 00 00 00 D1 00 00        |     |
| 000000h | A4 01 00 FF FF D2 00 00        |     |

| LDT   | LSB<br>00 01 02 03 04 05 06 07 | MSB |
|-------|--------------------------------|-----|
| 0140h | FF FF 00 00 00 D1 00 00        |     |
| 0138h | FE 00 00 01 FF D3 00 00        |     |
| 0130h | FF FF 00 F0 11 E1 00 00        |     |
| 0128h | FF FF 01 01 01 D2 00 00        |     |
| 0120h | FF FF 00 00 80 4F 00 00        |     |
| 0118h | FF FF AA 01 80 D2 00 00        |     |
| 0110h | FF FA 00 FF FF D2 00 00        |     |
| 0108h | EE 00 00 FF D2 00 00           |     |

| Page Directory | LSB<br>00 01 02 03 | MSB |
|----------------|--------------------|-----|
| Address        | Value              |     |
| 0000101Ch      | FF FF 00 00        |     |
| 00001018h      | FE 00 00 01        |     |
| 00001014h      | 01 00 00 F0        |     |
| 00001010h      | D0 00 01 01        |     |
| 0000100Ch      | FF 00 00 10        |     |
| 00001008h      | 03 00 00 20        |     |
| 00001004h      | A4 01 00 FF        |     |
| 00001000h      | 80 01 00 00        |     |

| Page Table 1 | LSB<br>00 01 02 03 | MSB |
|--------------|--------------------|-----|
| Address      | Value              |     |
| 20 00 00 1Ch | FF FF 00 00        |     |
| 20 00 00 18h | FE 00 00 01        |     |
| 20 00 00 14h | 01 00 00 F0        |     |
| 20 00 00 10h | D0 00 01 01        |     |
| 20 00 00 0Ch | FF 00 00 00        |     |
| 20 00 00 08h | 03 00 00 45        |     |
| 20 00 00 04h | 00 20 00 41        |     |
| 20 00 00 00h | 80 01 00 00        |     |

| Page Table 2 | LSB<br>00 01 02 03 | MSB |
|--------------|--------------------|-----|
| Address      | Value              |     |
| 10 00 00 1Ch | FF FF 00 00        |     |
| 10 00 00 18h | FE 00 00 01        |     |
| 10 00 00 14h | 01 00 00 F0        |     |
| 10 00 00 10h | D0 00 01 01        |     |
| 10 00 00 0Ch | FF 00 00 00        |     |
| 10 00 00 08h | 03 00 00 5C        |     |
| 10 00 00 04h | 00 20 00 1D        |     |
| 10 00 00 00h | 80 01 00 00        |     |

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**M.M: 40**

**13-05-2017**

**DURATION: 50min**

**Note: No partial marking in part A**

**Q1.** Arrange the following interrupts in 8086 microprocessor in **increasing** order of their priority:

A. INTO                      B. Single Step                      C. NMI                      D. INTR

[2M]

**Sol: D,C,B,A**

D C A B is the correct answer

**Q2.** Write down the **addressing mode** of the following instruction for an 8086 processor

[3M]

MOV AX,4020[BX+DI]

**Sol: Base relative plus index**

**Q3.** An 8086 processor is working with 10 MHz system clock, how much **time** (with proper units) one MEMR cycles takes if there are 4 wait states. [3M]

**Sol: ~~80ns~~**

800 ns

**Q4.** For an 8254 operating in mode-3 with input clock frequency of 2 MHz if the counter is loaded with 0000H then the output clock frequency is? (report up-to two decimals) [4M]

**Sol: 30.5175 Hz**

**Q5.** Determine the contents of **AL** register ( **in hex**) after completing execution of the following program snippet on 8086 processor. [4M]

MOV AL, 22H

MOV CL, 03H

ROL CL, CL

ROL AL, CL

**Sol: AL=22H**

**Q6.** Using **only logical** instructions (use minimum no of instruction) from 80286 instruction set, sign extend a 12-bit data in AX register into 16-bit data. Initially the upper nibble of AH is all zeros. [4M]

**Sol: ROL AX, 04**

**SAR AX, 04**

**SHL AH, 04**

**SAR AH, 04**

**Q7.** In 8086 write a sequence of four instructions that will set all the conditional flags and reset all the status flags. The flag register of 8086 is shown below. [4M]

|    |    |    |    |    |    |    |    |    |    |   |     |   |   |   |    |
|----|----|----|----|----|----|----|----|----|----|---|-----|---|---|---|----|
| 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5 | 4   | 3 | 2 | 1 | 0  |
| X  | X  | X  | X  | OF | DF | IF | TF | SF | ZF | X | ACF | X | P | X | CY |

**Sol: MOV AX, FFFFH**

**AND AX, 0700H**

**PUSH AX**

**POPF**

**Q8.** In 8086 the flag register is a 16-bit register in which the least significant bit of the upper byte

corresponds to trap flag. Write a sequence of instructions (code snippet) which can set the trap flag while leaving the other flags unchanged. [4M]

**Sol:**        **PUSHF**  
               **POP AX**  
               **OR AX, 0100H**  
               **PUSH AX**  
               **POPF**

**Q9.** Write an code snippet (need now show instruction for segment initialization) to generate a delay of 63ms using an 8086 processor running at 5MHz frequency. Use only instruction from the table given below: [4M]

| Instruction | No. of T-states needed |
|-------------|------------------------|
| MOV         | 4                      |
| NOP         | 3                      |
| DEC/INC     | 2                      |
| JNZ         | 16                     |

**Sol:**        **MOV CX, 3A98H / 15000D**  
               **DELAY: DEC CX**  
               **NOP**  
               **JNZ**  
               **DELAY**

**Q10.** An ISR is being is executed by 8086 and the contents of the stack at that time are shown below. If the ISR doesn't modify the contents of the stack during its execution then the return address (20-bit Hex) of the ISR is : [4M]

|     |                                                                                      |
|-----|--------------------------------------------------------------------------------------|
| 00H |  |
| 65H |                                                                                      |
| 00H |                                                                                      |
| 14H |                                                                                      |
| 00H |                                                                                      |
| 15H |                                                                                      |
| 08H |                                                                                      |
| 16H |                                                                                      |

**Sol: 1A500H**

**Q11.** Write the values of the destination after the execution of the code snippet listed below, write answer in hexadecimal only [4M]

```

STC
MOV AX, 5485H      ; AX = 5845H
RCR AL, 1          ; AL = C2H
MOV CL, 03         ; CL = 03H
RCL AX, CL         ; AX = A615H

```



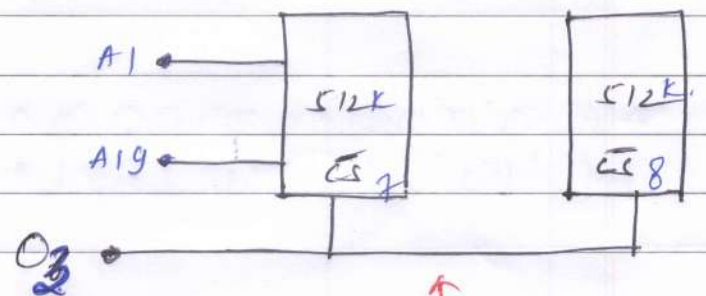
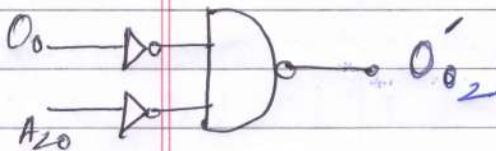
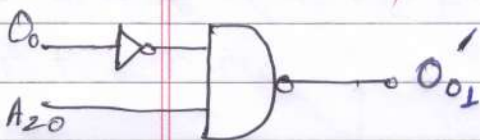
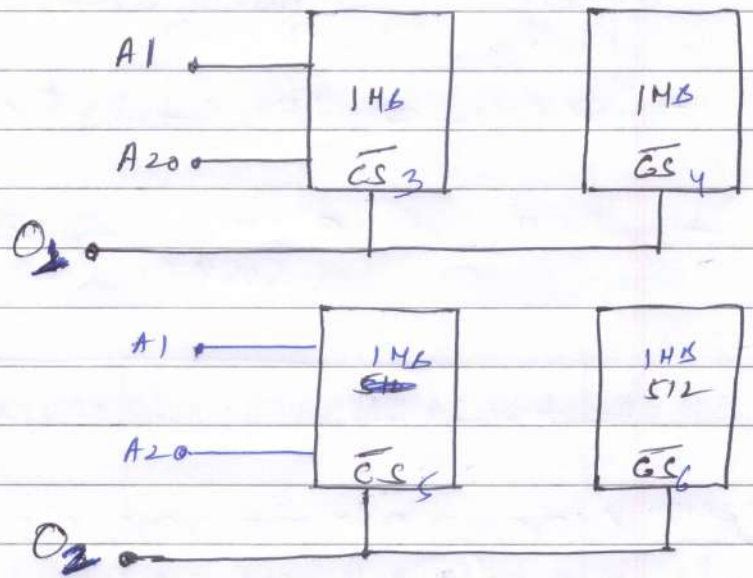
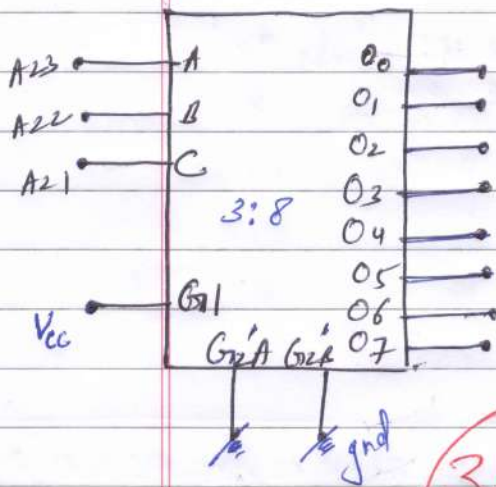
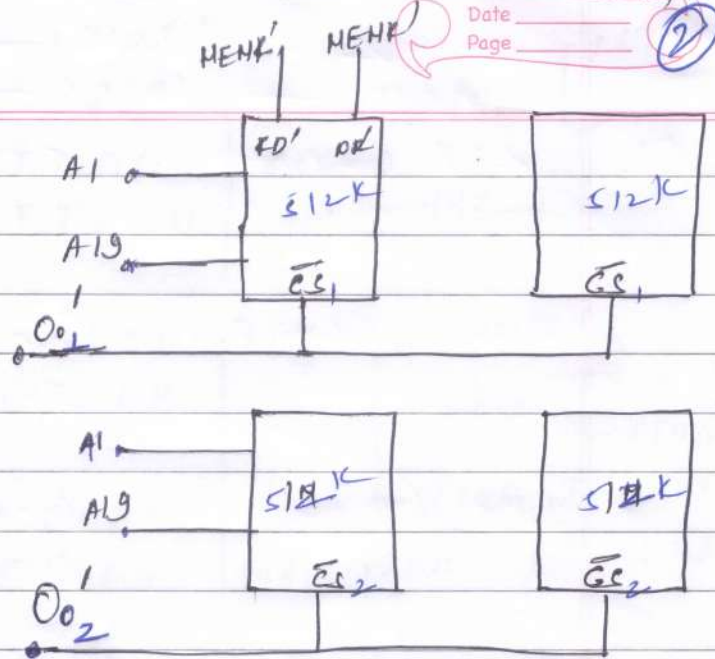
D<sub>15</sub> - D<sub>0</sub>  
Eren.

D<sub>7</sub> - D<sub>0</sub>  
classmate odd

Date  
Page

2

| 23 | 22 | 21 | 20 |                |
|----|----|----|----|----------------|
| 0  | 0  | 0  | 0  | O <sub>0</sub> |
| 0  | 0  | 0  | 1  |                |
| 0  | 0  | 1  | 0  | O <sub>2</sub> |
| 0  | 1  | 0  | 1  | O <sub>3</sub> |
| 0  | 1  | 1  | 0  | O <sub>6</sub> |
| 0  | 1  | 1  | 1  |                |



3M  
No partial marks

6M

- zero if decoding logic is wrong.

$32$   
 SRAM  $\rightarrow$  512 KB  $\rightarrow 2^{19}$   
 SRAM 2  $\rightarrow$  1 MB  $\rightarrow 2^{20}$   
 ROM  $\rightarrow$  512 KB  $\rightarrow 2^{19}$

classmate ①

Date  
Page

Q1.

2 MB (SRAM1) 0000000H  
1FFFFFFFH

2 MB

1 MB

512

512

512

4 MB (SRAM2) 2000000H  
5FFFFFFFH

4 MB

2 MB

1 MB

1 MB

2 MB

1 MB

1 MB

1 MB (ROM) 6000000H  
6FFFFFFFH

1 MB

512

512

- No of chips required: SRAM1  $\rightarrow \frac{2 \text{ MB}}{512 \text{ KB}} = 4 \text{ chips}$

- SRAM2  $\rightarrow \frac{4 \text{ MB}}{1 \text{ MB}} = 4 \text{ chips}$

- ROM  $\Rightarrow \frac{1 \text{ MB}}{512 \text{ KB}} = 2 \text{ chips}$

A23 A22 A21 A20 19 18 17 16 15 14 13 12 A11 A10 A9 A8 A7 A6 A5 A4 A3 A2 A1 A0

SRAM1 0000 0000 0000 0000 0000 0000 0000 0000 0000 0000  
0001 1111 1111 1111 1111 1111 1111 1111 1111 1111

SRAM2 0010 0000 0000 0000 0000 0000 0000 0000 0000 0000  
0101 1111 1111 1111 1111 1111 1111 1111 1111 1111

ROM 0110 0000 0000 0000 0000 0000 0000 0000 0000 0000  
0110 1111 1111 1111 1111 1111 1111 1111 1111 1111

6M only for correct hex



If the root directory entry of a file is the following

42 49 54 53 44 41 54 41- 58 4C 53 23 7B BD 4D 18

A4 3A 4A 3A 00 00 0F 20- CE 28 0F 00 04 00 00 04

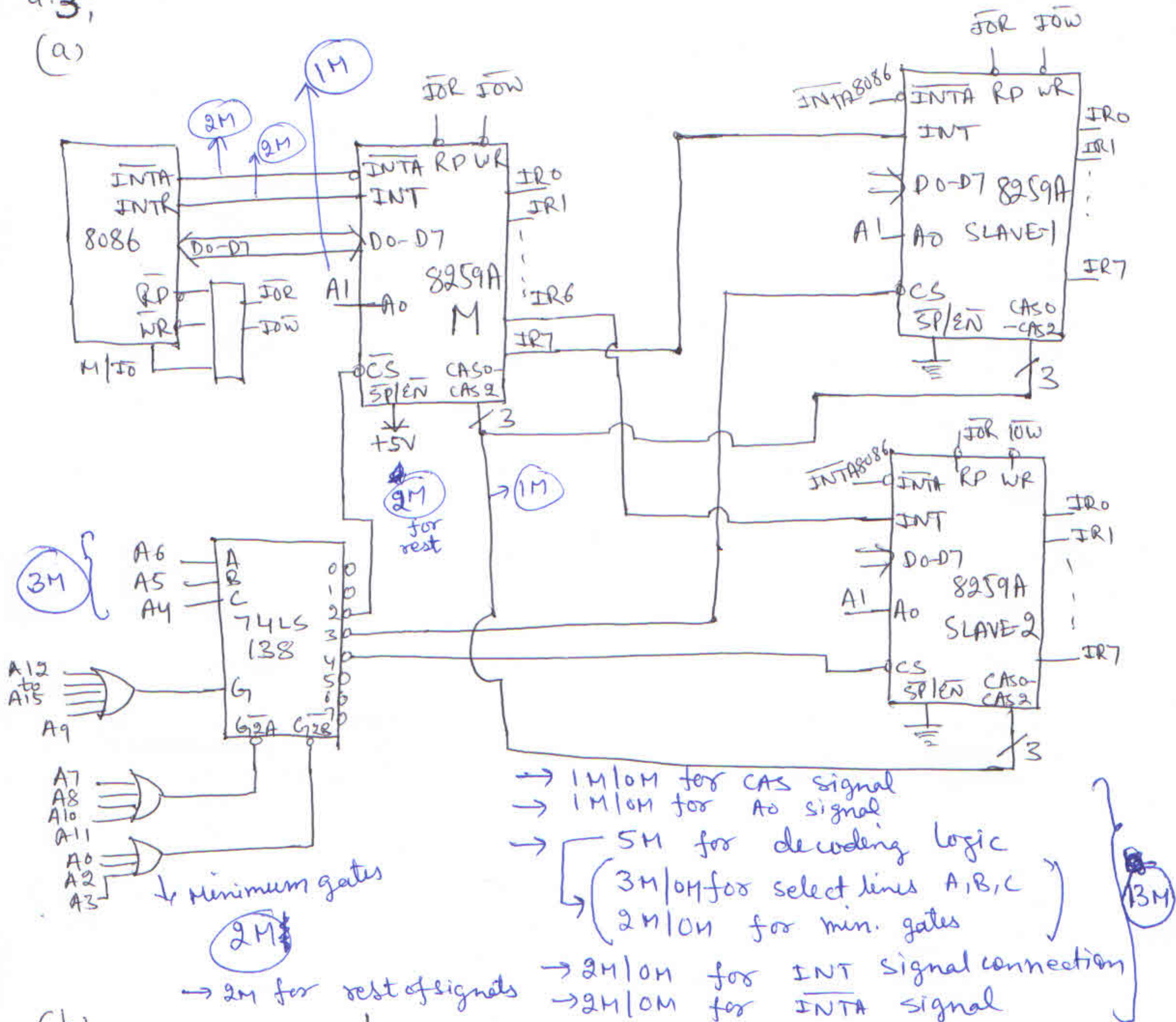
1. What are the attributes of this file?
2. What is the date of creation of this file?
3. What is the time of creation of this file?
4. What is the starting cluster number of this file?
5. What is the size of this file ?

Answers:

1. Hidden Read-only Archive file
2. 14<sup>th</sup> June 2000 or 2010
3. 04:00:30
4. 15<sup>th</sup>
5. 269168 bytes



Q.3,  
(a)



(b)

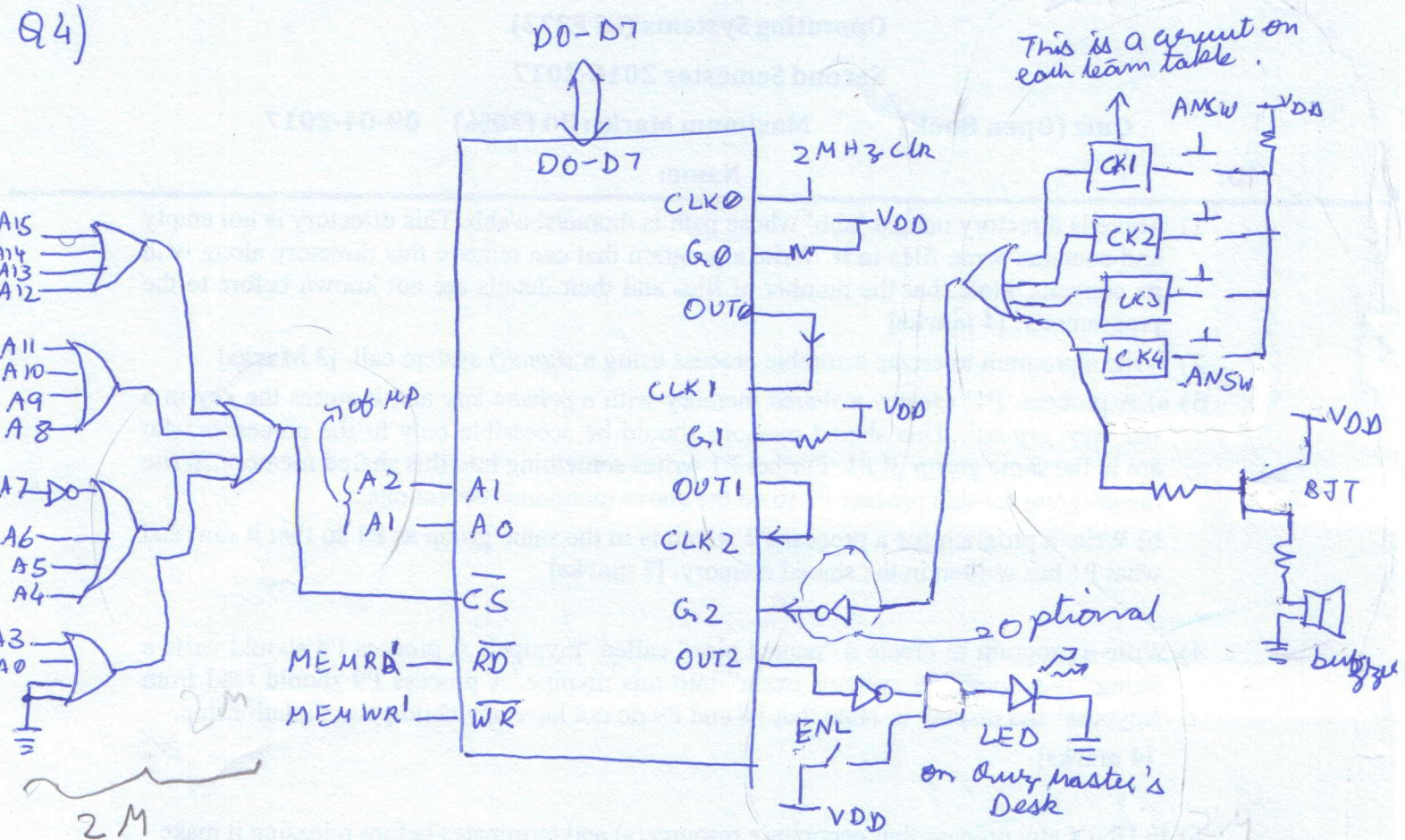
|      | 8259A-M | 8259A-Slave-1 | 8259A-Slave-2 |
|------|---------|---------------|---------------|
| ICW1 | 11      | 11            | 11            |
| ICW2 | 28      | F0            | F8            |
| ICW3 | C0      | 07            | 06            |
| ICW4 | 0F      | 0B            | 0B            |
| OCW1 | 3F      | 00            | 3C 08 00      |

IM/OM for each row

ON if values are not in Hex. Values are in Hex.



Q4)



Note: CK1 represents the circuit which produces pulse for team A

Note 2: Only one buzzer is shown here. Actually, there will be four such buzzers, Sufficient to show one buzzer.

Counter 2 interfacing  $\rightarrow 3M$   
Counter 0 and 1 interfacing  $\rightarrow 2M$

adds, underpinning  
and other signals  $\rightarrow 3M$

- b) Counter 0 Mode 3 or Mode 2 — 1M  
Counter 1 Mode 3 or Mode 2 — 1M  
Counter 3 Mode 5 only — 2M

Count in counter 0  $\frac{2 \times 10^6}{100} = (20,000)_{10} = 4B2014$

This provides a 100 Hz clock for counter 1



Count for counter 1 = 100 = 64H

This produces 1 Hz clock for counter 2

Count for counter 2 = 10 only = 0AH

This will produce a low strobe at end of 10 seconds

c)

MOV AL, 0011<sup>010</sup>0110b ; binary counter 0 mode 3

~~MOV~~ OUT 86H, AL ; control register prog

MOV AL, 0101<sup>010</sup>0110b

OUT 86H, AL

MOV AL, 10011010b ; counter 2 mode 5

OUT 86H, AL

MOV AL, 20H ; LSB into counter 0

OUT 80H, AL

MOV AL, 64H ; LSB into counter 1

OUT 82H, AL

MOV AL, 0AH ; LSB into counter 2

OUT 84H, AL

MOV AL, 4EH ; MSB into counter 0

OUT 80H, AL

2 marks each for each counter initialization and loading count

$$2 \times 3 = 6M$$



Q5.

**linear address as 00 80 10 00h**

**physical address as 41 00 20 00h. [8 + 8 Marks] BINARY MARKING**